1. Let P denote the plane spanned by $(1, 4, -1, -1)$ and $(1, 4, 1, 4)$ in $\mathbf{R}^4$

   1.1. Find a basis for $P^\perp$

   1.2. Find an implicit description of $P^\perp$

2. Consider the plane $2x - 3y + z = 0$ in $\mathbf{R}^3$

   2.1. Find a basis for its orthocomplement.

   2.2. Find an implicit description of its orthocomplement.

3. Let S denote the nullspace of the matrix below, find the bases for S and $S^\perp$.

$$\begin{bmatrix} 4 & 2 & 4 & 1 \\ -2 & 4 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

4. Find a basis for the orthocomplement of the line spanned by $(1, 1, 1, 1, 1) \subset \mathbf{R}^5$.

5. Find bases for each of the four subspaces associated with the given matrix below.

$$\begin{bmatrix} 2 & 4 & 4 & -2 \\ 1 & 2 & 4 & -3 \\ 1 & 2 & 3 & -2 \end{bmatrix}$$

6. Given the matrix $\mathbf{A} = \begin{bmatrix} 1 & 5 & -3 \\ 1 & -3 & 5 \end{bmatrix}$

   6.1. Find sets of vectors R and C that span $\text{row}(\mathbf{A})$ and $\text{col}(\mathbf{A})$, respectively.

   6.2. Find a set V such that $\text{row}(\mathbf{A}) = V^{\perp}$.

   6.3. What is the dimension of $\text{col}(\mathbf{A})^{\perp}$?